

Layered monoidal theories

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University College London

27 February 2026

Cambridge Logic and Semantics Seminar

Outline

Motivation

Syntax and semantics

Functor boxes and coboxes

Case studies

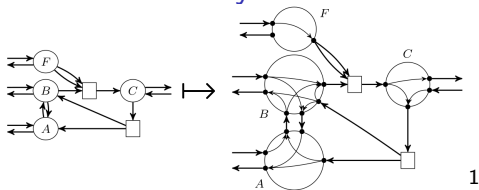
- Digital circuits

- Electrical circuits

- Chemical reactions

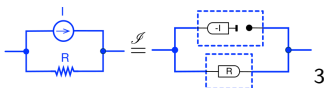
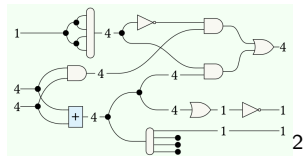
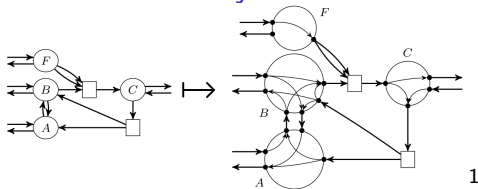
Related work

Motivation: layers of abstraction



¹ Andersen, Flamm, Merkle, Stadler. *Chemical Transformation Motifs*. 2017

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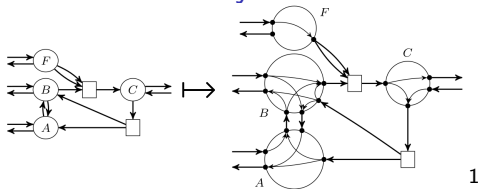


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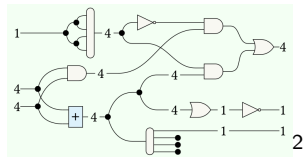
² Kaye. *Foundations of Digital Circuits*. 2024

³ Boisseau, Sobociński. *String Diagrammatic Electrical Circuit Theory*. 2021

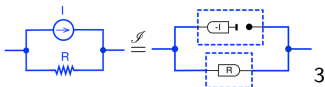
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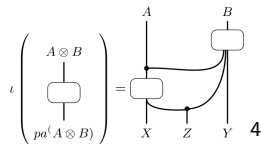
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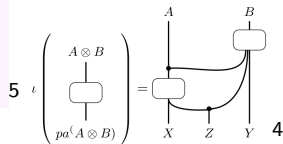
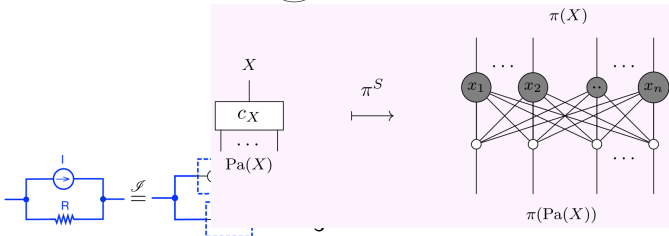
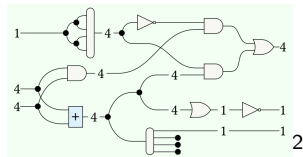
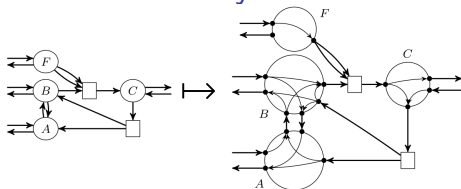
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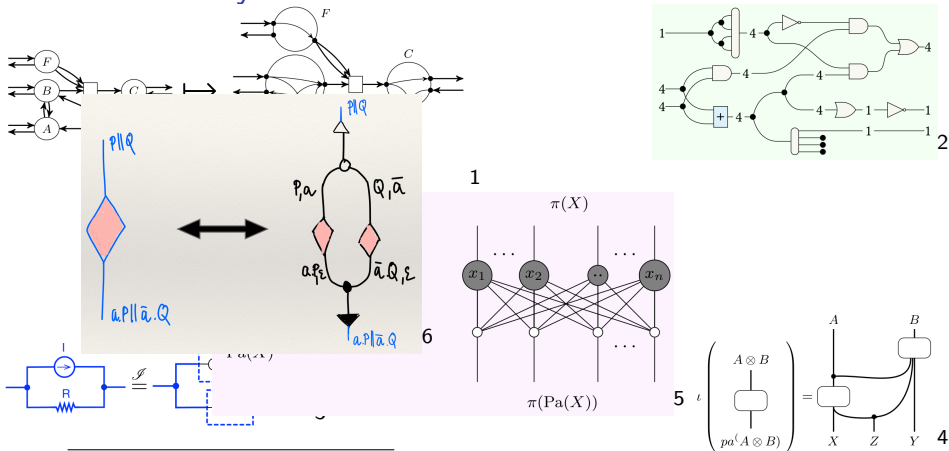
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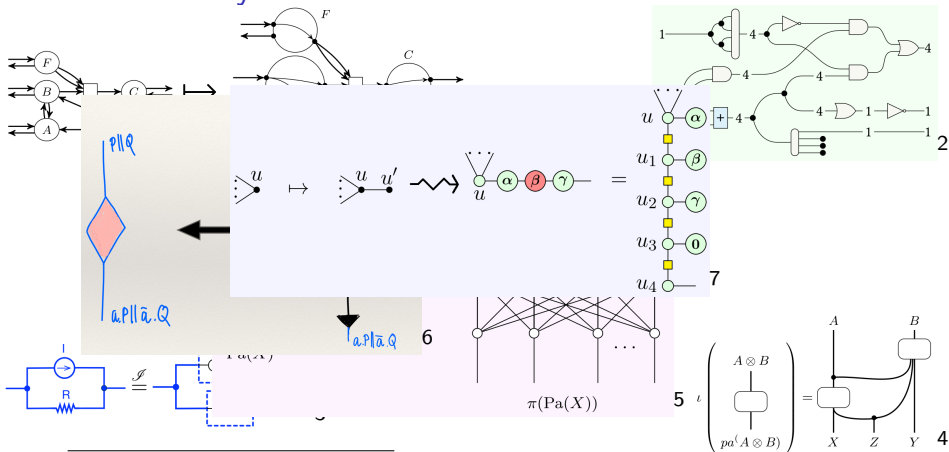
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⁷ Backens, Miller-Bakewell, de Felice, L., van de Wetering. *There and back again*. 2021

Signatures and types

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such that ε is a monoid with unit $\varepsilon : \varepsilon$.

Basic terms

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$$\begin{array}{c}
 \frac{\omega \in \Omega}{\boxed{} : (\varepsilon : \omega \mid \varepsilon : \omega)} \text{ (int-unit)} \quad
 \frac{A : \omega \quad A \neq \varepsilon}{\boxed{\text{---}} : (A : \omega \mid A : \omega)} \text{ (int-id)} \quad
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$$\begin{array}{c}
 \boxed{x} : (A : \omega \mid B : \omega) \quad f \in \mathcal{F}(\omega, \tau) \quad \boxed{x} : (A : \omega \mid B : \omega) \quad \boxed{y} : (C : \omega \mid D : \omega) \\
 \hline
 \boxed{f \boxed{x}} : (f(A) : \tau \mid f(B) : \tau) \quad \boxed{\boxed{x} \boxed{y}} : (AC : \omega \mid BD : \omega)
 \end{array}$$

(int-box) (int-tensor)

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$$\begin{array}{c}
 \frac{}{\boxed{} : (\varepsilon : \varepsilon \mid \varepsilon : \varepsilon)} \text{ (ext-unit)} \quad \frac{x : (T \mid S) \quad y : (S \mid U)}{x; y : (T \mid U)} \text{ (comp)} \quad \frac{x : (T \mid S) \quad y : (U \mid W)}{x \otimes y : (T, U \mid S, W)} \text{ (ext-tensor)}
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Theories

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- ▶ a set of 2-equations E^2 with respect to E^1 and η .

Structural equations

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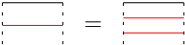
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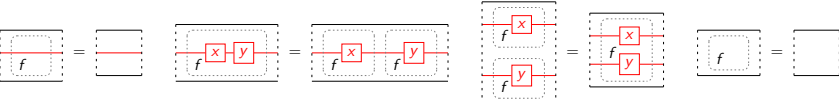
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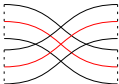
▶  .

Additional terms

Symmetry terms:

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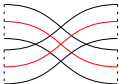


$: (A : \omega, B : \tau \mid B : \tau, A : \omega)$

The diagram shows two vertical dashed lines representing the boundaries of two sets of variables, A and B. Four curved lines connect the boundaries: two black lines and two red lines. The black lines represent the identity mapping (A to A, B to B), while the red lines represent the swap mapping (A to B, B to A). The red lines cross each other in the center, illustrating the swap operation.

Additional terms

Symmetry terms: $\frac{A : \omega \quad B : \tau}{\text{---}} \text{ (swap)}$



$: (A : \omega, B : \tau \mid B : \tau, A : \omega)$

Opfibrational terms:

Additional terms

Symmetry terms:

$$\frac{A : \omega \quad B : \tau}{\text{ (swap) }} \quad \text{Diagram: A box with two vertical dashed lines and two horizontal dashed lines. Two red curves cross each other, representing a swap.} : (A : \omega, B : \tau \mid B : \tau, A : \omega)$$

Opfibrational terms:

$$\begin{array}{l} \frac{A : \omega \quad f \in \Omega(\omega, \tau)}{\text{ (ext-gen) }} \quad \text{Diagram: A box with a horizontal red line and a vertical dashed line. An arrow labeled 'f' points to the left.} : (A : \omega \mid f(A) : \tau) \\[10pt] \frac{\omega \in \Omega}{\text{ (monoid-unit) }} \quad \text{Diagram: A box with a horizontal dashed line and a vertical dashed line. A red curve starts from the left and ends at the right.} : (\varepsilon : \varepsilon \mid \varepsilon : \omega) \\[10pt] \frac{A : \omega}{\text{ (diag-counit) }} \quad \text{Diagram: A box with a horizontal dashed line and a vertical dashed line. A red curve starts from the left and ends at the right, with a red dot at the end.} : (A : \omega \mid \varepsilon : \varepsilon) \\[10pt] \frac{A : \omega \quad B : \omega}{\text{ (monoid) }} \quad \text{Diagram: A box with two vertical dashed lines and two horizontal dashed lines. Two red curves start from the left and end at the right, representing a monoid multiplication.} : (A : \omega, B : \omega \mid AB : \omega) \\[10pt] \frac{A : \omega}{\text{ (diag) }} \quad \text{Diagram: A box with two vertical dashed lines and two horizontal dashed lines. A red curve starts from the left and ends at the right, with a red dot at the end, representing a diagonal.} : (A : \omega \mid A : \omega, A : \omega) \end{array}$$

Additional terms

Fibrational terms:

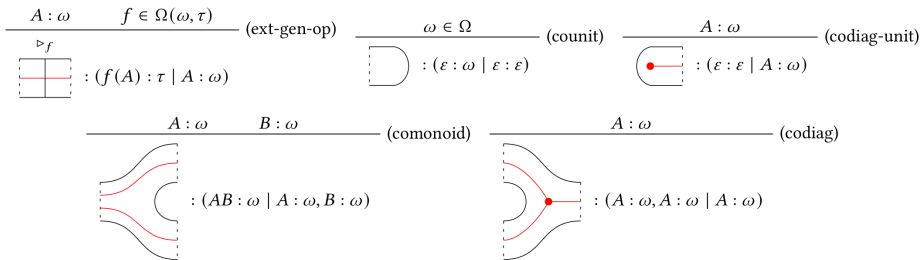
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 \begin{array}{ccc}
 \begin{array}{|c|c|} \hline \hline \hline \end{array} : (f(A) : \tau \mid A : \omega) & \begin{array}{|c|} \hline \hline \hline \end{array} : (\varepsilon : \omega \mid \varepsilon : \varepsilon) & \begin{array}{|c|} \hline \hline \hline \end{array} : (\varepsilon : \varepsilon \mid A : \omega)
 \end{array} \\
 \\
 \frac{A : \omega \quad B : \omega}{\text{ (comonoid)}} \quad \frac{A : \omega}{\text{ (codiag)}} \\
 \begin{array}{ccc}
 \begin{array}{|c|} \hline \hline \hline \end{array} : (AB : \omega \mid A : \omega, B : \omega) & & \begin{array}{|c|} \hline \hline \hline \end{array} : (A : \omega, A : \omega \mid A : \omega)
 \end{array}
 \end{array}$$

Additional terms

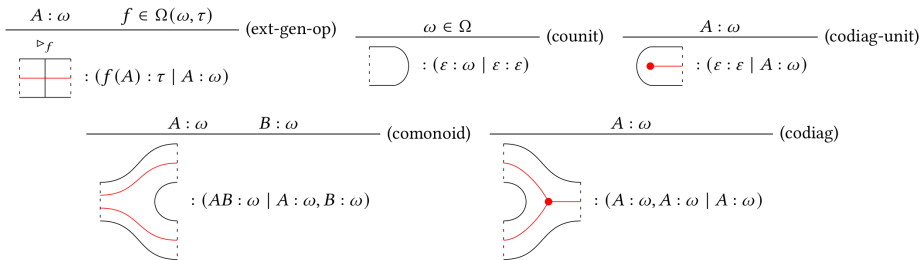
Fibrational terms:



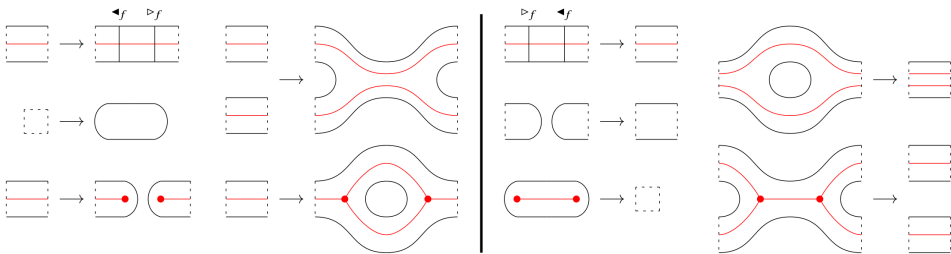
Structural 2-cells:

Additional terms

Fibrational terms:



Structural 2-cells:



Additional equations

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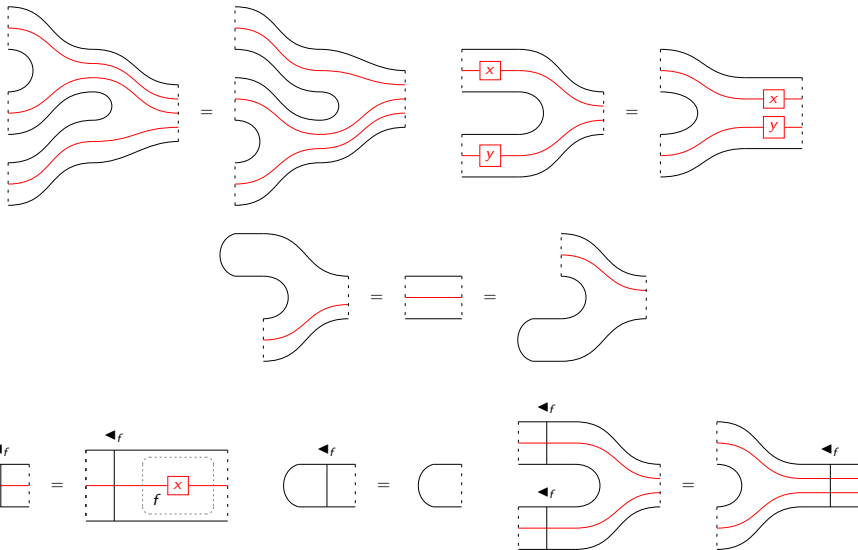
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 - ▶ Compatibility of 2-cells with sliding.

Additional equations



Semantics

theory	semantics
monoidal theory	(strict) monoidal category
opfibrational theory	split opfibration with indexed monoids
fibrational theory	split fibration with indexed comonoids
deflational theory	split monoidal deflation

Functor boxes

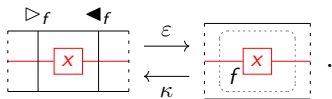
The usual notation for functor boxes decomposes as follows:

Functor boxes

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Proposition

In a deflational theory, there are bidirectional 2-cells

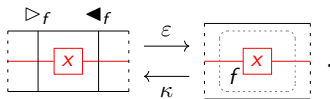


Functor boxes

The usual notation for functor boxes decomposes as follows:

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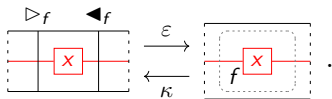
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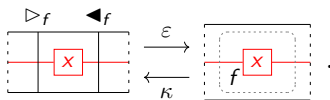
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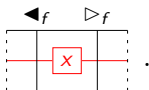
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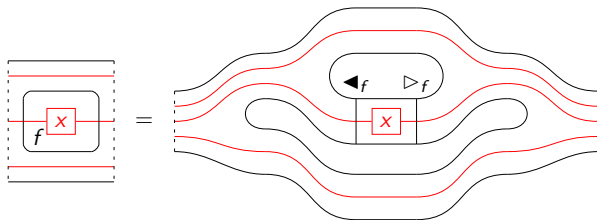
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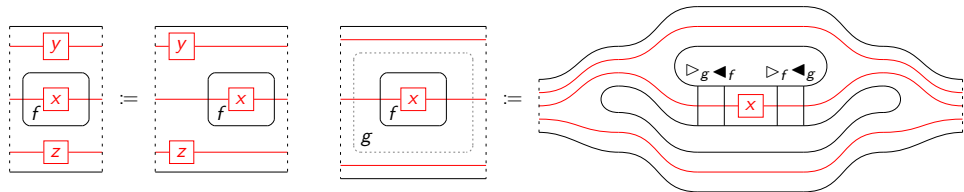
Note: NOT an internal term! However...

Functor coboxes

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In a deflational theory, there are bidirectional 2-cells (3) if and only if for all terms x and y with the sort $(A : \omega \mid B : \omega)$, the existence of 2-cells on the right of (4) implies the existence of 2-cells on the left of (4):

$$\begin{array}{c} \boxed{\boxed{f}} \rightleftarrows \boxed{\quad} \quad (3) \end{array} \quad \begin{array}{c} \boxed{\boxed{f \begin{array}{|c|} \hline x \\ \hline \end{array}}} \rightleftarrows \boxed{\boxed{f \begin{array}{|c|} \hline y \\ \hline \end{array}}} \Rightarrow \boxed{\boxed{x}} \rightleftarrows \boxed{\boxed{y}} . \quad (4) \end{array}$$

Functor coboxes

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$$\begin{array}{|c|} \hline \boxed{f} \\ \hline \end{array} \begin{array}{c} \longrightarrow \\ \longleftarrow \end{array} \begin{array}{|c|} \hline \\ \hline \end{array} \quad (3) \quad \begin{array}{|c|} \hline \boxed{f} \boxed{x} \\ \hline \end{array} \begin{array}{c} \longrightarrow \\ \longleftarrow \end{array} \begin{array}{|c|} \hline \boxed{f} \boxed{y} \\ \hline \end{array} \Rightarrow \begin{array}{|c|} \hline \boxed{x} \\ \hline \end{array} \begin{array}{c} \rightleftarrows \\ \rightleftarrows \end{array} \begin{array}{|c|} \hline \boxed{y} \\ \hline \end{array} . \quad (4)$$

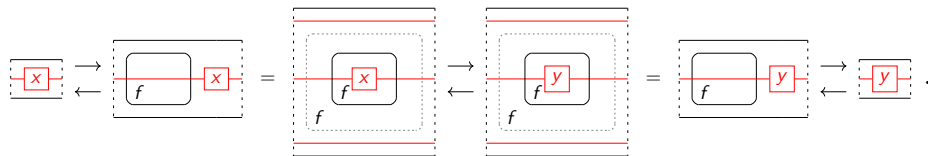
Moreover, if (3) is an equality, then the implication (4) holds also when the bidirectional 2-cells on both sides are replaced with equalities.

Functor coboxes

Proof.

Suppose the 2-cells (3) exist, and let x and y be terms such that $fx \rightleftharpoons fy$.

Then:

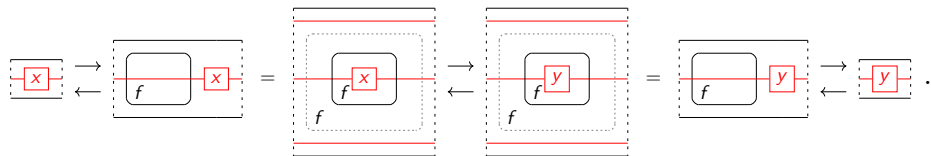


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Functor coboxes

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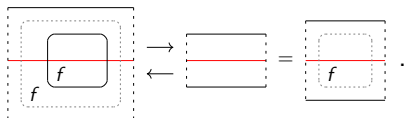
Suppose the 2-cells (3) exist, and let x and y be terms such that $fx \rightleftharpoons fy$. Then:



The same argument shows the case with equalities.

Conversely, suppose that the implication (4) holds. We use the following to

obtain the required 2-cells:



Digital circuits: Signature

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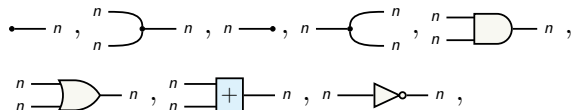
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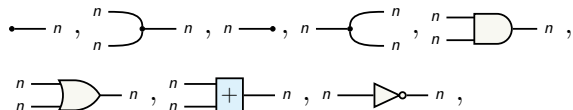
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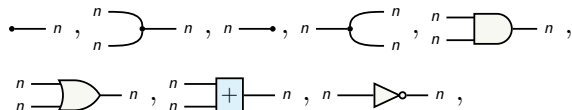


- ▶ for $n = 1$, the monoidal signature Σ_1 is as above, except that we replace the adder with $\begin{matrix} 1 & \text{---} & \boxed{+} & \text{---} & 1 \\ 1 & \text{---} & & & 1 \end{matrix}$.

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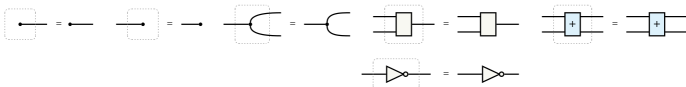


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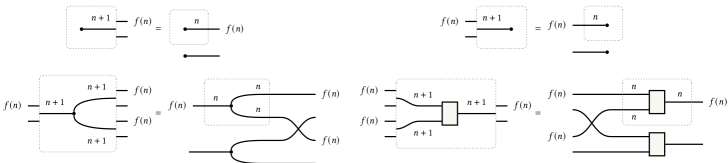
This signature is called *simple arithmetic circuits*.

Digital circuits: Theory

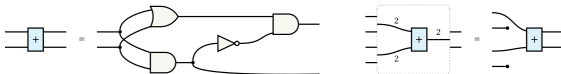
$$f \in \mathcal{F}(1, 1) : f(1) = 1$$



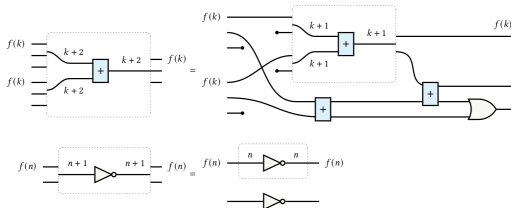
$$f \in \mathcal{F}(n+1, 1) : f(n+1) = f(n)1$$



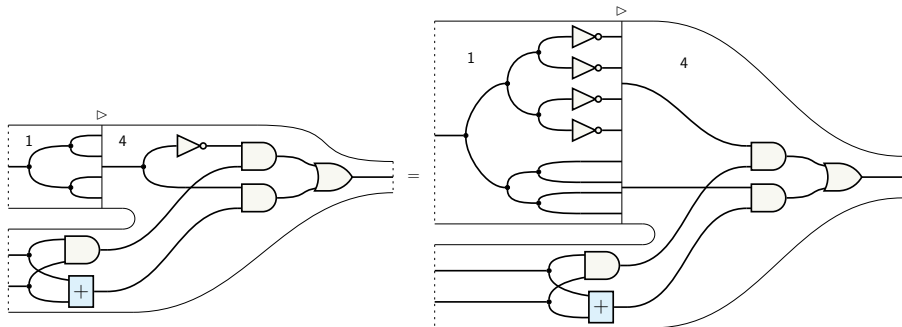
$$n = 1 :$$



$$n = k+1 \text{ for } k \geq 1 :$$



Digital circuits: A simple arithmetic logic unit



Electrical circuits: Signature

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$$\begin{array}{ccc} \mathbf{Bip} & \hookrightarrow & \mathbf{ECirc} \\ B \downarrow & & \downarrow \mathcal{I} \\ \mathbf{Imp} & \xrightarrow{W} & \mathbf{GAA}_{\mathbb{R}(x)} \end{array} \quad .$$

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The generators in **Imp** are given by all the terms in **GAA** with sort $(1, 1)$.
 The generators in **Bip** are given by the *bipoles* (5). The generators in **ECirc** are given by the bipoles (5) and (6).

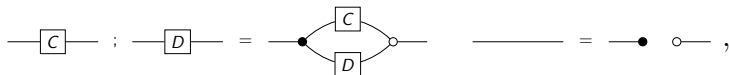


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The diagram shows an equation in the graphical affine algebra (GAA). On the left, a horizontal line passes through a square box labeled 'C', followed by a semicolon, then a horizontal line passing through a square box labeled 'D'. This is followed by an equals sign. To the right of the equals sign is a diagram where a horizontal line enters a black dot from the left, then splits into two parallel paths, each containing a square box labeled 'C' (top) and 'D' (bottom), which then merge at a white circle. This is followed by another equals sign. To the right of the second equals sign is a horizontal line entering a black dot from the left, followed by a white circle, and then a horizontal line exiting to the right.

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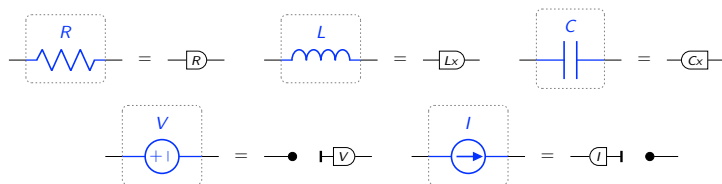
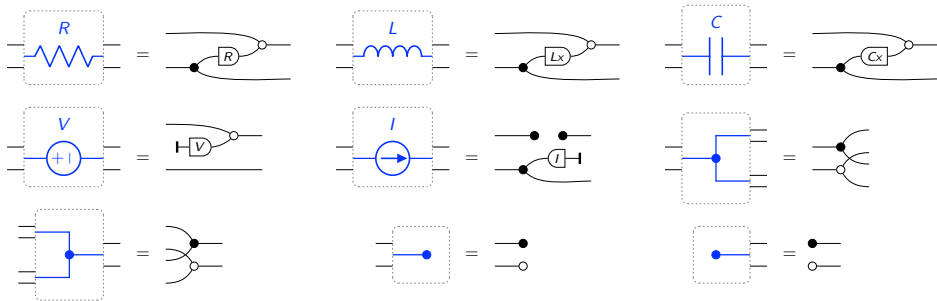
The diagram shows an equation in the Imp language. On the left, a horizontal wire passes through a box labeled 'C', followed by a semicolon, then another horizontal wire passing through a box labeled 'D'. This is followed by an equals sign. To the right of the equals sign is a diagram where a single input wire splits into two parallel wires, each passing through a box labeled 'C' and 'D' respectively, which then merge back into a single output wire. This is followed by another equals sign and a final diagram consisting of a solid black dot (multiplication) followed by a small open circle (addition) on a wire.

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The diagram shows an equation defining the functor W. On the left, a horizontal wire passes through a box labeled 'C' which is enclosed in a dashed rectangular border. This is followed by an equals sign. To the right of the equals sign is a diagram where a horizontal wire enters a box labeled 'C'. From the output of the box, the wire loops back to the input of the box, forming a feedback loop. The loop is completed by a small open circle at the top and a small solid black dot at the bottom.

- ▶ the functors $\mathcal{I} : \mathbf{ECirc} \rightarrow \mathbf{GAA}$ and $B : \mathbf{Bip} \rightarrow \mathbf{Imp}$ are defined on the next slide.

Electrical circuits: Theory



Electrical circuits: Impedance box

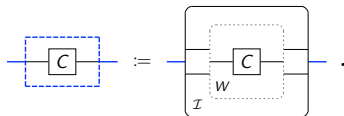
Within this layered theory, we recover the *impedance box* of Boisseau & Sobociński as a box-cobox combination, rather than an additional notational device.

Electrical circuits: Impedance box

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Definition

Let C be a generator of **Imp**. Define the *impedance box* as:



Electrical circuits: Impedance box

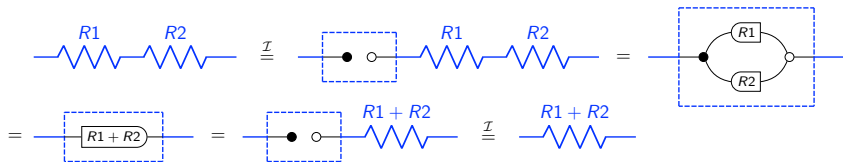
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Chemical reactions: Signature

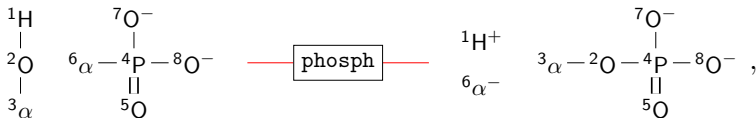
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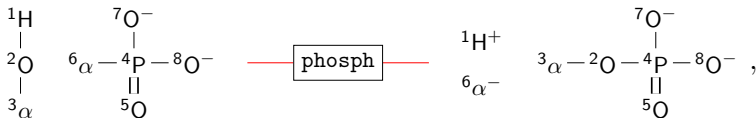
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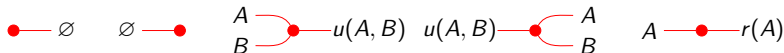
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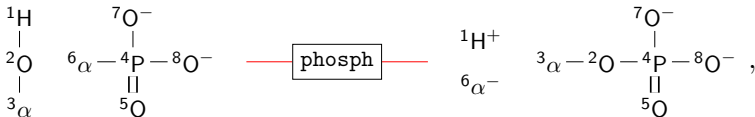
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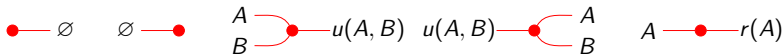
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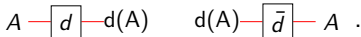
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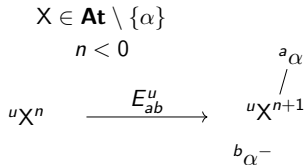
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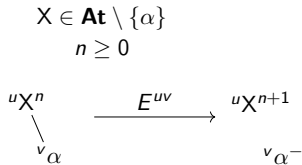
- ▶ the remaining generators in **Disc** are given by the *disconnection* rules



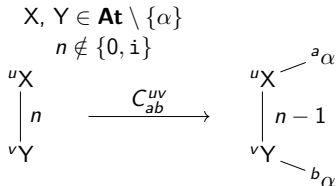
Chemical reactions: Disconnection rules



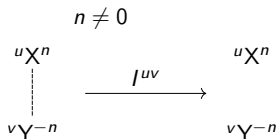
Electron detachment (negative charge)



Electron detachment (nonnegative charge)



Covalent bond breaking



Ionic bond breaking

Chemical reactions: Theory

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Chemical reactions: Theory

- ▶ the structural generators are isomorphisms in **Disc**,
- ▶ the functor **Scheme** $\rightarrow$ **Disc** is given by:

$$\boxed{\text{phosph}} = E^{46} C_{ab}^{12} E^{1a} \bar{E}^{4a} \bar{C}_{ab}^{42}$$

Chemical reactions: Theory

- ▶ the structural generators are isomorphisms in **Disc**,
- ▶ the functor **Scheme** $\rightarrow$ **Disc** is given by:

$$\boxed{\text{phosph}} = E^{46} C_{ab}^{12} E^{1a} \bar{E}^{4a} \bar{C}_{ab}^{42}$$

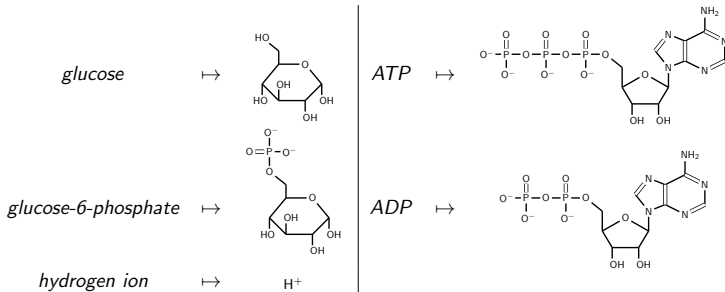
- ▶ the functor **Name** $\rightarrow$ **Disc** is defined below:

Chemical reactions: Theory

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- ▶ the functor **Name** $\rightarrow$ **Disc** is defined below:



Chemical reactions: Reaction

The reaction below is now derivable as a term in the layered theory (but not in any of the constituting monoidal theories):

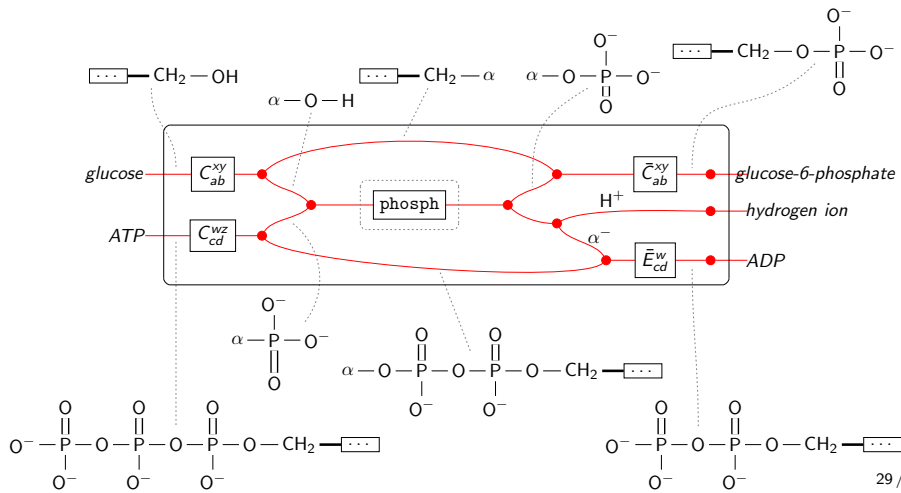
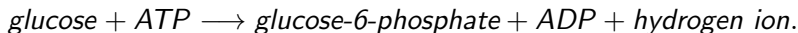
Chemical reactions: Reaction

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Related work

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Thank you for your attention!